Hints for RISS/GNSS Kalman Filter Project

1) You may use a Gauss-Markov model for gyro-bias error modelling. A Gauss-Markov Model is characterized by:
    a. Correlation time <tune this parameters from few minutes to few hours as needed>
    b. Standard Deviation <tune this parameter noting that this is a radians/sec unit>. A guide to this value may be estimated from stationary gyro-measurements

2) You may need an error state for error in odometer generated acceleration. You may model it as Gauss-Markov as well and tune it as described above.
3) For Q matrix Calculation: Based on the states of the filter, a Q matrix element would be tuned for each state for best performance.
4) For Measurements standard deviation (R matrix), you may use the incoming GPS measurements standard deviation scaled by a factor. This factor can be also tuned for best performance as well.

An example for R matrix calculation:

```
lat_std_in_meters = BGP_Lat_std(k);       %k is the GPS epoch
long_std_in_meters = BGP_Long_std(k);     %k is the GPS epoch

N = a./sqrt(1-e2*sin(BGP_Lat(k)*D2R).*sin(BGP_Lat(k)*D2R));% D2R:deg to rad
M = a*(1-e2)./((1-e2*sin(BGP_Lat(k)*D2R).*sin(BGP_Lat(k)*D2R)).^(1.5));

lat_std_in_rad = lat_std_in_meters/(M+BGP_Alt(k));
long_std_in_rad = long_std_in_meters/((N+BGP_Alt(k))*cos(BGP_Lat(k)*D2R));

R_(1,1) = (lat_std_in_rad)^2;
R_(2,2) = (long_std_in_rad)^2;
R_(3,3) = (BGP_Alt_std(k))^2;
R_(4,4) = (long_std_in_meters*5)^2;
R_(5,5) = (lat_std_in_meters*5)^2;
R_(6,6) = (BGP_Alt_std(k)*10)^2;

R_ = R_* Scale;
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